

Course: B. Tech.

Branch :Civil

Semester :III

Subject Code & Name: BTCVC 305 Surveying

Max Marks: 60

Date:

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q. 1 Solve Any Two of the following.

- A) What is meant by combined corrections in leveling? Derive its equation. CO1 6
- B) Differentiate between radiation & intersection. CO2 6
- C) A dumpy level was setup & leveled with its eyepiece over a peg C. The height from top of the center of eyepiece was measured & found to be 1.246 m. The staff reading on peg D was 0.892m. The instrument was then moved & set up at D. The height of eyepiece above D was 1.362m. The reading on staff held on peg C was 1.720 m. Determine the true Reduce level of the peg D if that of peg C was 150m. CO3 6

Q.2 Solve Any Two of the following.

- A) A contour plan is to be prepared for building site. State the factors you will consider for the choice of contour interval CO1 6
- B) What are various methods of contouring? Explain any one with neat sketch. CO2 6
- C) The following observations were made during testing of dumpy level. CO3 6

Instrument at	Staff reading at	
	A	B
A	1.725	2.245
B	2.145	3.045

What should be R.L. of B if R.L. of A is 450.00m?

Q. 3 Solve Any One of the following.

- A) What is zero circle? What is need of finding the area of zero circle? CO1 6
- B) Write note on Sensitivity of bubble tube. CO2 6
- C) What is orientation of plane table? Explain any one method of it. CO3 6

Q.4 Solve Any Two of the following.

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| A) What do you understand by the term Balancing the traverse? State various rules available for this process. | CO1 | 6 |
| B) Explain the procedure for carrying out preliminary survey for proposed irrigation canal. | CO2 | 6 |
| C) Explain spire test for adjusting the Transit theodolite. | CO3 | 6 |

Q. 5 Solve Any One of the following.

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| A) Explain with neat sketch the method of transferring the alignment inside a tunnel. | CO1 | 6 |
| B) Write a note on Principle of EDM. | CO2 | 6 |
| C) The following perpendicular offsets were taken at 10m intervals from a survey line to an irregular boundary line 3.82, 4.37, 6.82, 5.26, 7.59, 8.90, 9.52, 8.42 & 6.43m. Calculate the area enclosed between survey line & boundary by-i) Simpson's Rule ii) Trapezoidal Rule | CO3 | 6 |

***** End *****